



# Emden: A novel method integrating graph and transformer representations for predicting the effect of mutations on clinical drug response

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## ARTICLE INFO

### Keywords:

Mutation effect prediction  
Clinical drug response  
Deep learning  
Interpretable model

## ABSTRACT

Precision medicine based on personalized genomics provides promising strategies to enhance the efficacy of molecular-targeted therapies. However, the clinical effectiveness of drugs has been severely limited due to genetic variations that lead to drug resistance. Predicting the impact of missense mutations on clinical drug response is an essential way to reduce the cost of clinical trials and understand genetic diseases. Here, we present Emden, a novel method integrating graph and transformer representations that predicts the effect of missense mutations on drug response through binary classification with interpretability. Emden utilized protein sequences-based features and drug structures as inputs for rapid prediction, employing competitive representation learning and demonstrating strong generalization capabilities and robustness. Our study showed promising potential for clinical drug guidance and deep insight into computer-assisted precision medicine. Emden is freely available as a web server at <https://www.psymukb.net/Emden>.

## 1. Introduction

In recent decades, precision medicine has been developed for treating numerous diseases [1,2]. However, the clinical efficacy of molecular-targeted drugs has been frequently limited by mutations that affect the ligand binding to proteins, which causes drug resistance [3–7]. Initial characterization of individual genetic information allows us to prioritize clinical treatment strategies and identify or design more appropriate drugs as part of a new personalized medicine [8,9]. In the past, identifying genotypes related to drug responsiveness required time-consuming and expensive *in vitro* or *in vivo* functional studies [10]. Therefore, there is a need for an alternative computational method that enables rapid and accurate identification of drug-resistance mutations.

Previous studies have laid the groundwork for using computational prediction methods to quantify changes in ligand affinity resulting from mutations [3,11–15]. In these studies, the mutation-induced impairment of drug sensitivity was defined as a difference in the protein-drug binding free energy ( $\Delta\Delta G$ ) between wild-type and mutated target

proteins, such as mCSM-lig [3], MutationFEP [13] and PremPLI [14]. However, there is currently no established threshold or standard to determine the clinical manifestation that corresponds with  $\Delta\Delta G$ . Moreover, categorizing the effect of mutations on drugs by category would provide a more straightforward and simplified approach to clinical assistance. The reported classification tools, DRSP<sup>1</sup><sub>s</sub> and DRSP<sup>1</sup><sub>c</sub>, explored the relationship between mutations and cancer drug resistance from the perspective of protein structures [1]. However, the dependence of crystal structures on protein and docking complex hinders their rapid application, while the tailored design for anti-cancer drugs somewhat restricts their scope of application.

In this study, we present a deep-learning method, Emden, to predict the effect of missense mutations on clinical drug response through binary classification with interpretability. We processed and merged two manually collected clinical annotated datasets, PharmGKB [16] and CIViC [17], as the benchmark of our work. A deep learning network consisting of a graph convolutional network (GCN) [18–20] and transformer encoders [21,22] was utilized to perform representation

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learning. Emden only utilized protein sequences-based features and drug structures as inputs, which enabled its ability for unstinted prediction. We observed that Emden exhibited robust performance when inputting reverse mutations, and it also demonstrated competitive generalization and predictive capabilities through external validation and tool comparison. In summary, our approach provided insights into the effect of mutations on drug sensitivity, indicating its potential for clinical applications in precision medicine and drug development. All the services as well as the standalone tool of Emden are freely accessible at the web server <https://www.psymukb.net/Emden>.

## 2. Materials and methods

### 2.1. Benchmark dataset

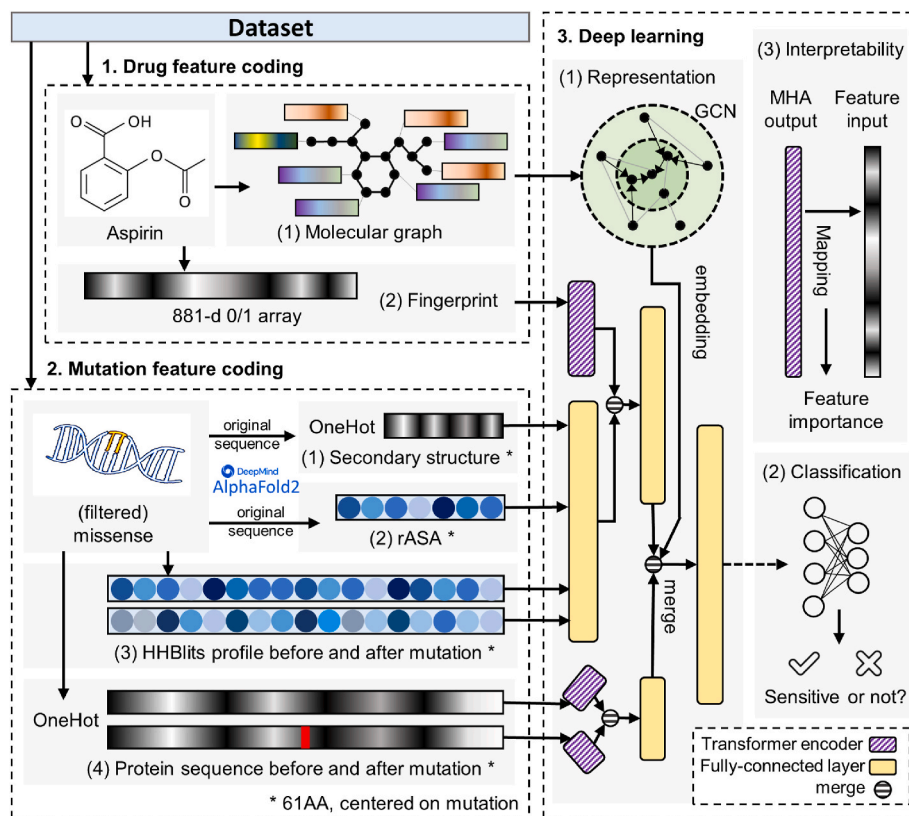
In this study, we combined two clinical evidence datasets, PharmGKB [16] and CIViC [17], to increase the number of available annotated clinical evidence records. PharmGKB curators manually create clinical annotations by consolidating variant annotations that pertain to the same variant and drug phenotype [23], while CIViC is a community-driven database that utilizes expert crowdsourcing to interpret clinical variants in cancer CI [17]. Both datasets were accessed in their December 21st, 2022 versions. A series of preprocessing steps were undertaken to filter the data, including deduplication and conflict resolution (see [Supplementary Table 1](#)). Following preprocessing, 710 binary drug response evidences were retained, each of which contained a single-point missense mutation. The processed dataset exclusively consists of small molecule drugs, all of which are cataloged in PubChem [24]. From this set, 130 evidences were randomly selected to form an independent test set, with the remaining evidences used to train five cross-validation models. [Supplementary Table 2](#) provides a detailed description of the class distribution of our dataset across each fold.

### 2.2. Feature engineering and model architectures

Emden is a deep-learning network specifically designed as a two-class classification predictor to predict the clinical impact of missense mutations on drug response, based solely on the input of the molecular structure of the drug and primary protein sequences.

As depicted in [Fig. 1](#), the drug coding branch utilized a graph convolutional network (GCN) [18] to encode the SMILE [25] of each drug into a molecular graph. Meanwhile, the PubChem fingerprint [26] was processed by a transformer encoder [21]. Regarding the protein sequence coding branch, we employed AlphaFold2 [27] and DSSP [28] to generate the secondary structure (SS) and relatively accessible surface area (rASA) of the original sequence, while the HHBlits software [29] was used to create the HHBlits profile of both the original and mutated sequence. Several studies have suggested that directly feeding the spatial coordinates into the model, which represents the positions of protein residues, may not be the optimal approach [30,31]. One potential explanation for this is that deep learning networks might struggle to directly correlate atomic coordinates with the protein's folding pattern. Therefore, the residue coordinates were not used as input features. All of the features mentioned above were then inputted into a fully-connected layer. Additionally, we encoded the original and mutated protein sequences using the One-Hot coding strategy [32] and subsequently fed them into two separate transformer encoders, respectively. All sequence features were experimentally sliced into 61 amino acids, centered on the mutation position, to balance prediction accuracy with computational cost.

After concatenating intermediate abstract features generated by fully-connected layers of varying sizes, the advanced abstract features are ultimately fed through a fully-connected layer with a SoftMax activation function [33] to produce the final prediction output. In this work, the interpretability of the deep learning model is provided by the



**Fig. 1.** The model architecture and workflow of Emden. Only the molecular structure of drugs and the primary protein sequences were taken as the pipeline's input. The prediction output showed whether a drug was sensitive to a missense mutation. The interpretability of Emden is provided by the self-attention layer contained in the transformer encoder.

self-attention layer contained in the transformer block, and we interpret the importance of different input features by mapping the attention output to the corresponding dimensions of the input features.

### 2.3. Implementation details

The Emden model was implemented, trained, and tested using the Pytorch [34] open-source software library on an Nvidia GeForce RTX 3090 GPU. A batch size of 128 and an initial learning rate of 0.0005 were selected for training. We trained the network to minimize the cross-entropy loss function to classify the inputs. The model weights were initialized using the default settings in Pytorch, and an Adam optimizer [35] was used to train the parameters in all cases. The Transformer encoder of the drug input branch contained eight attention heads, while each protein sequence input branch contained four attention heads.

### 2.4. Performance evaluation

As our study focuses on two-class classification for clinical drug response prediction, we evaluated the performance of our predictor using the following metrics [36]: accuracy, recall, precision, F1-score, and AUC, which were defined as:

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$F1\text{-score} = 2 \times \frac{Recall \times Precision}{Recall + Precision}$$

where TN, TP, FN, and FP represent the number of correctly predicted non-sensitive mutations, the number of correctly predicted sensitive mutations, the number of falsely predicted non-sensitive mutations, and the number of falsely predicted sensitive mutations, respectively. The overall success of our test was primarily assessed using the area under the ROC curve (AUC), which was defined as

$$AUC = \int_0^1 \Pr[TP](v)dv$$

where  $\Pr[TP]$  is a function of  $v = \Pr[FP]$ . A poor test has an  $AUC \approx 0.5$ , while a good test has an  $AUC \approx 1$ .

## 3. Results

### 3.1. Performance and robustness of Emden

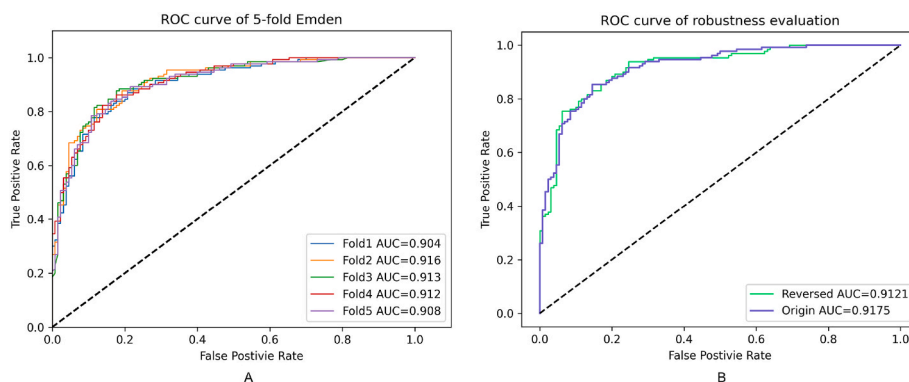
The comprehensive prediction performance of Emden was evaluated using 5-fold cross-validation on the training dataset. As illustrated in Fig. 2A and Supplementary Table 3, Emden achieved an average prediction AUC of 0.91, ACC of 0.84, and F1-score of 0.88 on the independent test set. Additionally, we also demonstrated the superior learning ability of Emden by comparing its performance to six widely used traditional machine learning methods with grid search, such as Naïve Bayes [37] and random forest [38,39] (see Supplementary Fig. 1, Supplementary Tables 3–4, with each value of the input treated as an independent feature for machine learning).

Given the inherent tendency for mutations to occur, machine-learning models may exhibit deviations after training [40–43]. To address this potential issue, we conducted a further analysis by exchanging the input features of the wild-type protein and the mutated protein on the independent test set, with the labels remaining unchanged. The final model, trained on the entire training set, was utilized to generate predictions for both the reversed and original test sets. As expected, the prediction performances on the two test sets were found to be nearly identical (see Fig. 1B and Supplementary Table 5), demonstrating that Emden was not affected by the direction of the mutation but rather made predictions based on information such as the presence of amino acid changes in a protein sequence, indicating its robustness.

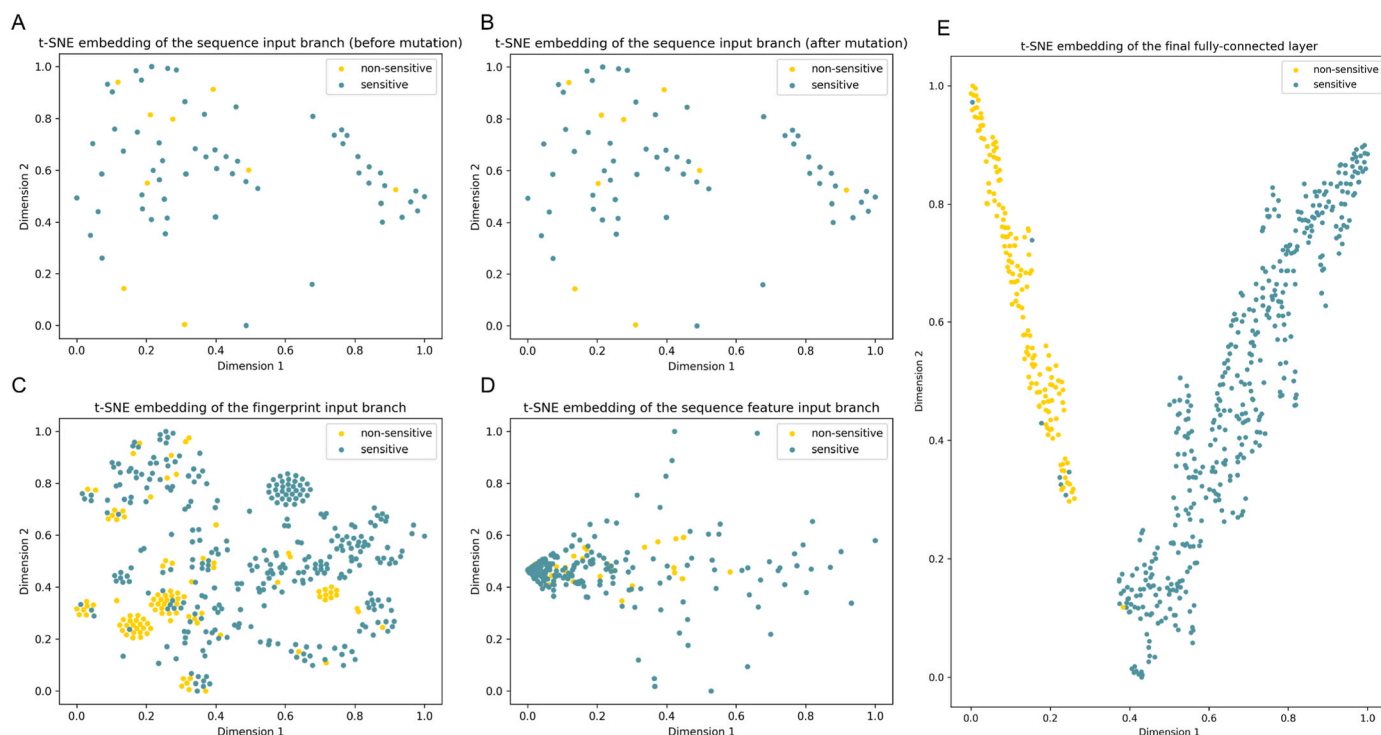
### 3.2. Representation and ablation study

To investigate the representation power of Emden and its capacity to extract high-level features from the input data, we used t-SNE [44] to visualize the initial features of the four input branches and the final features prior to the output layer on the entire training dataset. As illustrated in Fig. 3A–D, no discernible meaningful clusters were observed in the initial features of the four input branches. In contrast, the two clusters representing the two categories displayed a distinct classification boundary in Fig. 3E, indicating that Emden could identify an effective classification hyperplane for the training dataset.

As the model incorporated four input branches (the smile-GCN branch, PubChem fingerprint branch, the protein profile branch, and the OneHot sequence branch), a sequential ablation approach was employed to evaluate the necessity of each branch individually, whereby one branch was removed at a time. We conducted each ablation experiment by retraining the model and applying 5-fold cross-validation. The results of the ablation test, as shown in Table 1, indicated that each input branch had a distinct contribution, although there was some overlap between them. Furthermore, among these four input



**Fig. 2.** The prediction performance and robustness of Emden. (A) The ROC curve of 5-fold cross-validation performance on the independent test set. Each model was trained on the 5-fold training set. (B) The ROC curve of Emden on the original test set and the reversed test set. These two evaluations were performed using the model trained on the entire training set.



**Fig. 3.** The t-SNE visualization results show the representation power of Emden. (A) t-SNE embedding of the wild-type protein sequence input branch. (B) t-SNE embedding of the mutated protein sequence input branch. (C) t-SNE embedding of the PubChem fingerprint input branch. (D) t-SNE embedding of the sequence profile (SS + rASA + HHblits) input branch. (E) t-SNE embedding of the final fully-connected layer.

**Table 1**

The 5-fold cross-validation ablation study of Emden. The model comprised four input branches, with each branch evaluated for its necessity by removing it in turn.

| Model                                      | ACC                       | Precision                 | Recall                    | F1-score                  | AUC                       |
|--------------------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Emden                                      | <b>0.8354</b><br>± 0.0078 | <b>0.8557</b> ±<br>0.0177 | <b>0.9080</b><br>± 0.0218 | <b>0.8807</b><br>± 0.0056 | <b>0.9104</b><br>± 0.0091 |
| Without smile-GCN                          | 0.8077<br>± 0.0223        | 0.8416 ±<br>0.0096        | 0.8782<br>± 0.0421        | 0.8506<br>± 0.0196        | 0.8905<br>± 0.0073        |
| Without PubChem fingerprint                | 0.7646<br>± 0.0143        | 0.8064 ±<br>0.0179        | 0.8552<br>± 0.0474        | 0.8289<br>± 0.0154        | 0.8422<br>± 0.0085        |
| Without HHblits profile                    | 0.7786<br>± 0.0564        | 0.7906 ±<br>0.063         | 0.9040<br>± 0.0495        | 0.8576<br>± 0.0209        | 0.8767<br>± 0.0203        |
| Without AF2-generated profiles (rASA + SS) | 0.8153<br>± 0.0206        | 0.8376 ±<br>0.0503        | 0.9080<br>± 0.0586        | 0.8681<br>± 0.0122        | 0.8954<br>± 0.0133        |
| Without sequence OneHot                    | 0.8138<br>± 0.0178        | 0.8368 ±<br>0.0179        | 0.8989<br>± 0.0551        | 0.8653<br>± 0.0180        | 0.8825<br>± 0.0147        |

branches, we found that the PubChem fingerprint and sequence profile had a more significant impact on prediction.

### 3.3. Attention interpretability

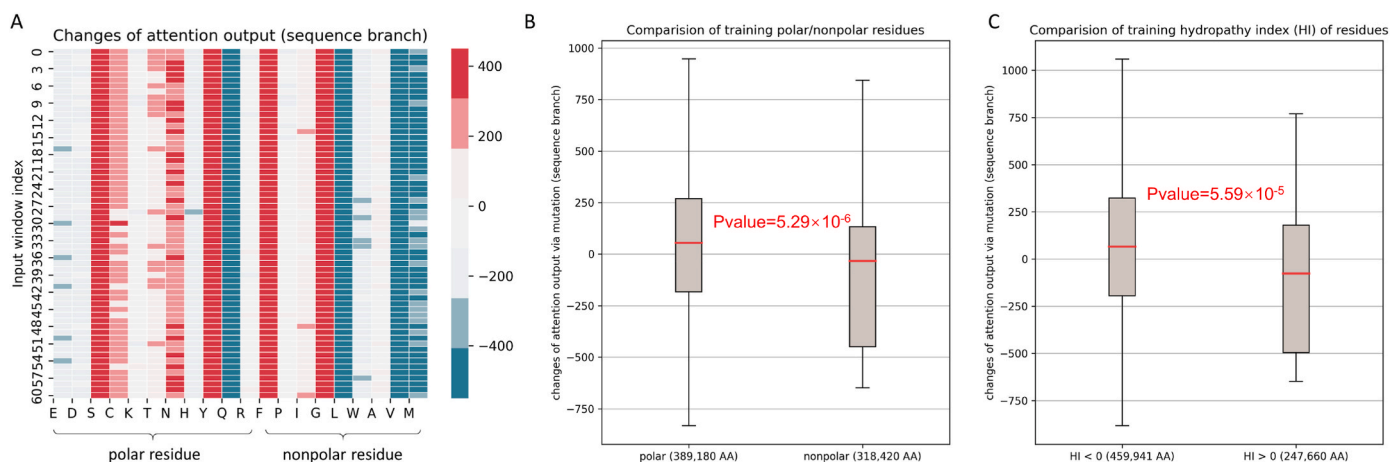
It is reported that drug-resistant variants are often surrounded by amino acid residues with neutral and low polarizability properties, which implies that the effectiveness of drugs may be influenced by the polarizability of these residues [1]. To investigate whether Emden captured such kinds of underlying information from the protein sequences that enhance prediction accuracy, we mapped the self-attention output of both wild-type and mutated sequence branches onto the input features. For each input sample in the training set, we subtracted the

mutated attention output from the original attention output to show the attention difference between these two OneHot sequence inputs. Then, the results of each sample were aligned based on the input window dimension and summed up. As is shown in Fig. 4A and Supplementary Table 6, the output of polar and nonpolar amino acids [45] tended to have two distinct distribution patterns, the changes in the attention output of polar residues tended to be positive while the output of nonpolar residues tended to be negative, indicating that the model made a distinction between these two types of amino acids during the process of learning feature representation, and considered polarity information to be an important factor influencing model classification.

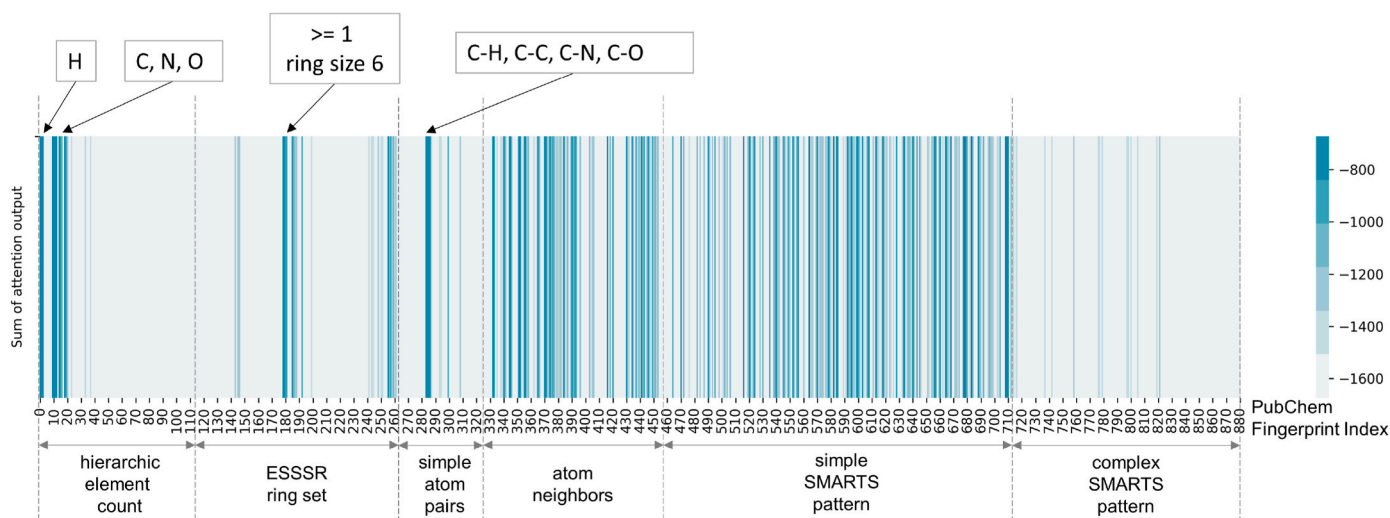
We further summarized the outputs of two types of residues according to their polarity and conducted a Mann-Whitney rank-sum test [46]. The results showed a statistically significant contribution difference between these two types of residues ( $P = 5.29 \times 10^{-6}$ ), which confirmed the conclusion stated earlier (Fig. 4B). We then repeated the same statistical analysis on the hydrophilicity of amino acids [47] and also found a significant difference in the distribution between the two categories (Fig. 4C and Supplementary Table 6,  $P = 5.59 \times 10^{-5}$ ). This suggested that hydrophilicity may also play an important role in the prediction.

The PubChem fingerprints denote chemical substructures defined in the PubChem database, which aid in similarity searching and neighboring, and are frequently employed for predicting drug-target interactions [48]. As for the PubChem fingerprint input branch, we aggregated the attention output of each sample along the PubChem fingerprint index and by aligning and summing them up (Supplementary Table 7). In Fig. 5, each index represents whether the current drug contains a specific substructure. The darker color of an index indicated a significant difference in attention output compared to the surrounding indices with lighter colors. It was observed that the model gave a more special response to the dark-colored index area and passed on this information to subsequent layers of the network, such as hydrogen atoms, carbon atoms, benzene rings, atom neighbors, and some simple patterns. The discrimination value of the attention output indicated that Emden





**Fig. 4.** The change of self-attention output between the wild-type and mutated protein sequence input branch. (A) the output of polar and nonpolar amino acids tends to distribute in two different directions, with most output of polar residues  $> 0$  and nonpolar residues  $< 0$ . (B) Comparison between the attention output of polar and nonpolar residues. The Mann-Whitney test results show a statistically significant difference between these two types of residues ( $P = 5.29 \times 10^{-6}$ ). (C) Comparison between the attention output of polar and nonpolar residues. The Mann-Whitney test results also show a statistically significant difference between these two types of residues ( $P = 5.59 \times 10^{-5}$ ).



**Fig. 5.** The self-attention output of the PubChem fingerprint branch. The PubChem system generates a binary substructure fingerprint for chemical structures. A fingerprint is an ordered list of binary bits (1/0), and each bit represents a Boolean determination of, or test for, the presence of a substructure (Supplementary Table 7). The darker color of an index indicated a significant difference in attention output compared to the surrounding indices with lighter colors.

successfully utilized PubChem fingerprint as a valuable feature and identified a suitable representation of the fingerprint for subsequent prediction.

### 3.4. Comparison of existing methods with external anti-cancer drug resistance evidence

In recent decades, targeted cancer therapy has significantly improved survival rates and become a standard strategy for treating cancer [49]. However, drug resistance in chemotherapy remains a problem [50]. Therefore, predicting the mutation effects of cancer drugs in advance can effectively reduce clinical trial costs and aid in developing personalized treatment strategies to resolve this issue. To evaluate the performance of Emden in predicting anti-cancer drug resistance, we compared it with two existing anti-cancer drug resistance predictors, DRSP<sub>s</sub><sup>i</sup> and DRSP<sub>c</sub><sup>i</sup> [1]. Since there were limited entries available for validation that did not overlap with the data used in our study, we utilized the same test set reported by DRSP (Table 2). The three test cases used for this comparison analysis were not included in the training or the

**Table 2**

Comparison between Emden and the other two methods on the test set of DRSP. The three evidences used for comparison were not included in the training or the test set of Emden.

| Protein | Variant | Drug        | True Label | DRSP <sub>s</sub> <sup>i</sup> Result | DRSP <sub>c</sub> <sup>i</sup> Result | Emden Result |
|---------|---------|-------------|------------|---------------------------------------|---------------------------------------|--------------|
| BRAF    | L505H   | Vemurafenib | 0          | 1                                     | 0                                     | 0            |
| MAP2K2  | V215E   | PD0325901   | 0          | 1                                     | 0                                     | 0            |
| ROS1    | G2032R  | Crizotinib  | 0          | 0                                     | 0                                     | 0            |

test set of Emden. The input feature of DRSP<sub>s</sub><sup>i</sup> consisted of a simulated structure of the drug-protein binding complex, whereas DRSP<sub>c</sub><sup>i</sup> utilized experimentally measured crystal complexes [1].

As shown in Table 2, Emden achieved correct prediction results on all three test cases, with accuracy exceeding DRSP<sub>s</sub><sup>i</sup> and comparable to DRSP<sub>c</sub><sup>i</sup> that used experimentally determined structural information. Without relying on real protein structure or docking complex, Emden

demonstrated its generalization ability as effectively predict the response of anti-cancer drugs to mutations.

3.5. Case studies

Anaplastic lymphoma kinase (*ALK*), also referred to as *CD246*, is a transmembrane tyrosine receptor kinase that belongs to the superfamily of insulin receptors, which plays a crucial role in both normal development and oncogenesis [51]. *ALK* can lead to cancer through the formation of a combined gene *NPM-ALK* or by increasing DNA replication or duplication [52]. Here, we selected six test cases about *ALK* on the independent test set to analyze the effectiveness of Emden (Table 3). Brigatinib, Crizotinib, Alectinib, and Ceritinib were anti-cancer drugs and acted as inhibitors of *ALK* [52]. We observed that Emden accurately predicted almost all evidence, except for an incorrect judgment on L1196M-Alectinib. It is noteworthy that when focusing on G1202R-Brigatinib vs. V1180L-Brigatinib, despite the different labels of these two pieces of evidence, Emden still made correct predictions about the effects of different mutations on the same drug. Similar accurate judgments were also reflected in cases where the same mutation acted on different drugs, such as V1180L-Ceritinib vs. V1180L-Crizotinib. The results of this case indicated that Emden considered both mutation and drug information during the prediction process, which reflected the effectiveness of the training process.

Another case was about a widely studied SNP, V158M, of Catechol-O-methyltransferase (*COMT*). *COMT* is vital in conjugating catechol estrogens generated during estrogen metabolism [53], and V158M plays a vital role in intelligence, personality, and logical memory [54]. Table 4 demonstrated that Emden accurately predicted the effects of V158M on the clinical response of Clozapine, Modafinil, and Paroxetine. These case studies showed the effectiveness of Emden in prediction and its potential for clinical applications.

4. Discussion

The emergence of drug resistance caused by mutations will have an impact on clinical treatment. Accurately predicting drug-resistant mutations would reduce trial and error costs, helping to achieve precision medicine more effectively. In this study, we developed a deep learning-based tool called Emden that predicted the impact of the mutation on drug response through binary classification using clinically annotated evidence. Emden did not require any protein crystals or docking complexes, with only mutation sequence and drug structure used for unrestricted prediction. It was demonstrated that Emden has competitive representation learning and generalization ability with robustness, and the visualization of attention output also provided interpretability to the model. In summary, due to the aforementioned advantages, Emden showed promising potential for clinical drug guidance, drug development, and screening of existing drugs for novel applications. We also provided a web server for users to conveniently explore our method.

Some limitations can still be found in Emden. The study faced a primary impediment due to insufficient clinical annotated evidence, which made it difficult to balance the fitting degree and the generalization ability of the model. However, as the volume of data increases in the future, Emden's architecture will aid in achieving more stable prediction performance. Notably, accurate prediction of drug response influenced by mutations is challenging since there are subtle differences between missense mutations. Furthermore, the training and prediction of the model are restricted solely to the available reported evidence, which essentially relies on statistical inference, thereby rendering it challenging to ascertain the errors that may arise due to interindividual differences. The prediction performance of Emden on multi-point mutations was not evaluated as well.

It is worth noting that predicting the effect of missense mutations accurately is challenging because of the subtle differences between them. As deep learning continues to rapidly advance, it is highly

Table 3

A case of *ALK*-related evidence on the independent test set.

| Gene       | Variant | Drug       | True Label | Emden Result | Source |
|------------|---------|------------|------------|--------------|--------|
| <i>ALK</i> | G1202R  | Brigatinib | 0          | 0            | CIViC  |
|            | G1202R  | Crizotinib | 0          | 0            |        |
|            | L1196M  | Alectinib  | 1          | 0            |        |
|            | V1180L  | Brigatinib | 1          | 1            |        |
|            | V1180L  | Ceritinib  | 1          | 1            |        |
|            | V1180L  | Crizotinib | 0          | 0            |        |

Table 4

A case of V158 M of *COMT* on the independent test set.

| Gene        | Variant | Drug       | True Label | Emden Result | Source   |
|-------------|---------|------------|------------|--------------|----------|
| <i>COMT</i> | V158M   | Clozapine  | 1          | 1            | PharmGKB |
|             |         | Modafinil  | 1          | 1            |          |
|             |         | Paroxetine | 1          | 1            |          |

probable that forthcoming methodologies will exhibit greater accuracy in predicting the impact of mutations on clinical drug response. In the future, we will continue to enhance Emden by tracking and collecting more clinically reliable data and investigating more sophisticated deep-learning structures. With the advancement of large language models (LLMs), we will evaluate the model performance when utilizing output representations from LLMs as feature coding strategy. We are also considering other representation models, such as enhanced heterogeneous graph convolution network as it has strong capacity for integrating multiple omics data [55]. What's more, we aim to broaden Emden's application scope to include RNA and DNA sequences, as viruses undergo rapid mutations as well.

Code availability

The source code of Emden is available at <https://github.com/Liuzh30/Emden>.

Funding

This work was supported by grants from STI 2030—Major Projects (no. 2022ZD0209100), the National Natural Science Foundation of China (nos. 81971292, 82150610506, and 62372099), the Natural Science Foundation of Shanghai (no. 21ZR1428600), the Medical-Engineering Cross Foundation of Shanghai Jiao Tong University (nos. YG2022ZD026 and YG2023ZD27), SJTU Trans-med Awards Research (no. 20220103), and Jilin Scientific and Technological Development Program (Nos. 20230401092YY, 20230201090GX, and 20210101175JC).

Declaration of competing interest

The authors declare no competing interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compbimed.2023.107678>.

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